optronix®

PLATINUM SERIES

Technical Data Sheet

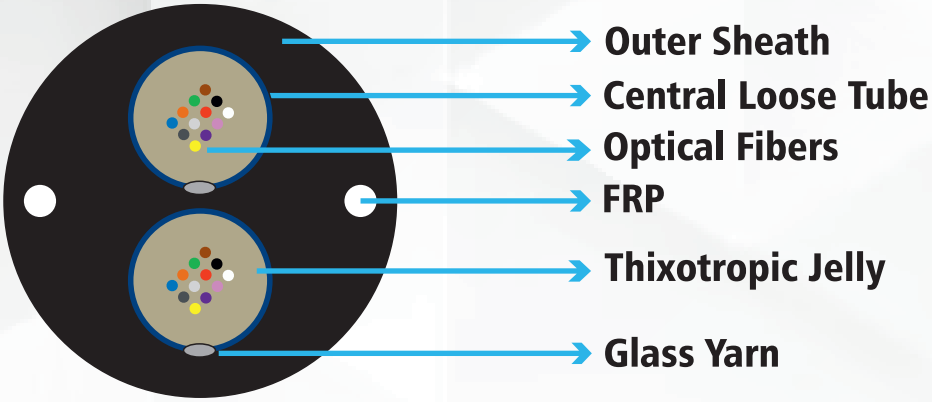
DUAL TUBE SM 24F

Un-Armoured OPTICAL FIBRE CABLE

Date: 01.09.2025

Construction Details	
No of Fiber and Color	: 24F per Tube - Blue, Orange, Green, Brown, Slate, White, Red, black, Yellow, violet, Pink, Aqua, Blue* Orange*, Green*, Brown*, Slate*, White*, Red*, Natural*, Yellow*, Violet*, Aqua*. (Where will be Single Ring Marking Fiber)
Loose Tube	: PBT Loose Tube Filled With Thixotropic Jelly
No of Loose Tube	: 2 Loose Tube
Tube Identification	: Natural or Any Colour
Strength Member	: 2 Nos. -1.0 mm UV Coated FRP Embedded in Sheath 7 Glass Yarn Over Loose Tube
Outer Jacket	: HDPE UV Black Colour

Schematic Diagram Not to Scale



Fibre Characteristics (As per ITU-T Rec. G. 652 D)

Attenuation (Transmission Characteristics)		**Geometrical Characteristics**	
@ 1310 nm	: ≤ 0.4 (dB/Km)	Mode Field Diameter @ 1310 nm	: 9.2 ± 0.4 μm
@ 1550 nm	: ≤ 0.2 (dB/Km)	Mode Field Diameter @ 1550 nm	: 10.4 ± 0.5 μm
		Cladding Diameter	: 125 ± 1 μm
Dispersion		Cladding Non Circularity	: ≤ 1%
A. Total Dispersion (Chromatic Dispersion)		Core Clad Concentricity Error	: ≤ 0.5 μm
1285-1330nm	: ≤ 3.5 ps/nm.km	Coating Diameter(Uncolored)	: 245 ± 10 μm
1270-1340nm	: ≤ 5.3 ps/nm.km	Coating/Cladding Concentricity	: ≤ 12 μm
1550nm	: ≤ 18 ps/nm.km	**Cut Off Wavelength**	
1625nm	: ≤ 22.0 ps/nm.km	Fiber Cut Off Wavelength	: ≤ 1320 nm
B. Polarization Mode Dispersion at 1310 & 1550 nm		Cable Cut Off Wavelength	: ≤ 1260 nm
At Fibre Stage	: ≤ 0.2 ps/sqrt.km	**Mechanical & Operating Characteristics**	
At Cable Stage	: ≤ 0.3 ps/sqrt.km	Operating Temperature	: -20° C to +70° C
C. Dispersion Slope & Wave Length		Fibre Proof Test	: 1%
Zero Dispersion Wavelength	: 1300-1324 nm	Stripability Force	: 1.3 < F < 8.9 N
Zero Dispersion Slope	: ≤ 0.092 ps/nm².km	Fibre Curl	: ≥4 meter radius of curvature

Cable Mechanical & Physical Characteristics

Cable Mechanical Characteristics		**Cable Physical Characteristics**	
Max. Tensile Strength	: 600 N (Short Term) Fiber Strain 0.6%	Cable Diameter	: 6.8mm(-0.1, +0.2)mm
Crush Resistance	: 1000 N/10 CM (Short Term)	Nominal Cable Weight	: 40kg/km ± 5%
Impact Resistance	: 5 N-m	Packing Length	: 2 km ± 2%
Min. Bending Radius (Long Term)	: 20D (D= Cable Diameter mm)	Packing Details	: In Coil

All Tests Shall be Carried out as per IEC Standards. Change in attentuation Shall be ≤0.05 dB/km

Cable Printing Details: (Hot Yellow Emb.)	OFC wwY OPTRONIX PLATINUM xxF CFRP1 Yw24 DT SM G652D Month YYYY Length Code Meter Marking Note: ww will be replaced with outer diameter of 58, 60, 65, 70. xx will be replaced with Fiber Count 24.

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PLATINUM SERIES

Technical Data Sheet

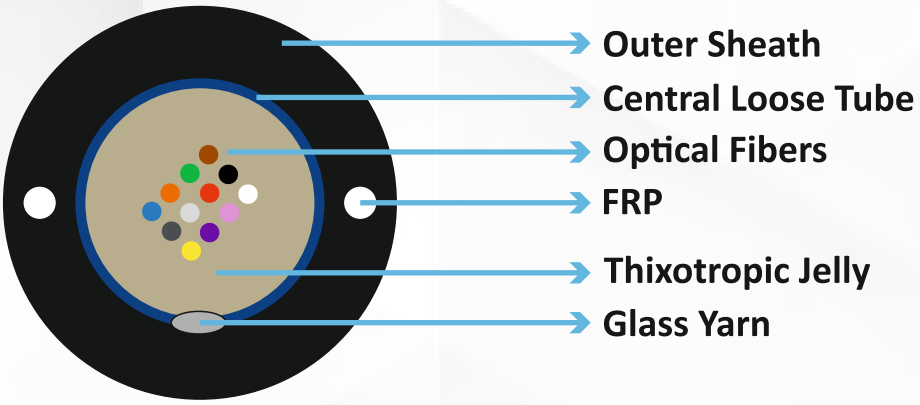
UNITUBE SM 2F, 4F, 6F, 12F G.652D

Un-Armoured OPTICAL FIBRE CABLE

Date: 24.04.2023

Construction Details	
No of Fiber and Color	: 02F per Tube - Blue, Natural.
	: 04F per Tube - Blue, Orange, Green and Natural.
	: 06F per Tube - Blue, Orange, Green,Brown, Gray and Natural.
	: 12F per Tube - Blue, Orange, Green, Brown, Gray, White, Red, Black, Yellow, Violet, Pink and Natural.
FRP	: UV-Cured Coated FRP
Loose Tube	: PBT Loose Tube Filled With Thixotropic Jelly
HDPE Coating	: Virgin HDPE
Lining Option	: (i) L1 Black Line on Orange Cable (ii) L2 Orange Line on Black Cable
Yarn Count	: 1200 tex
No of Loose Tube	: 1 Loose Tube
Tube Identification	: Natural
Strength Member	: 2/3 Nos. (0.8mm / 1mm) - FRP Embedded in Sheath
Outer Jacket	: PE Black/Orange Color

Schematic Diagram Not to Scale



Fibre Characteristics (As per ITU-T Rec. G. 652 D)

Attenuation (Transmission Characteristics)		**Geometrical Characteristics**	
@ 1310 nm	: ≤ 0.4 (dB/Km)	Mode Field Diameter @ 1310 nm	: 9.2 ± 0.4 μm
@ 1550 nm	: ≤ 0.2 (dB/Km)	Mode Field Diameter @ 1550 nm	: 10.3 ± 0.5 μm
		Cladding Diameter	: 125 ± 1 μm
Dispersion		Cladding Non Circularity	: ≤ 1%
A. Total Dispersion (Chromatic Dispersion)		Core Clad Concentricity Error	: ≤ 0.5 μm
1285-1330nm	: ≤ 3.5 ps/nm.km	Coating Diameter(Uncolored)	: 245 ± 10 μm
1270-1340nm	: ≤ 5.3 ps/nm.km	Coating/Cladding Concentricity	: ≤ 12 μm
1550nm	: ≤ 18 ps/nm.km	**Cut Off Wavelength**	
1625nm	: ≤ 22.0 ps/nm.km	Fiber Cut Off Wavelength	: ≤ 1320 nm
B. Polarization Mode Dispersion at 1310 & 1550 nm		Cable Cut Off Wavelength	: ≤ 1260 nm
At Fibre Stage	: ≤ 0.2 ps/sqrt.km	**Mechanical & Operating Characteristics**	
At Cable Stage	: ≤ 0.3 ps/sqrt.km	Operating Temperature	: -20° C to +70° C
C. Dispersion Slope & Wave Length		Fibre Proof Test	: 1%
Zero Dispersion Wavelength	: 1300-1324 nm	Stripability Force	: 1.3 < F < 8.9 N
Zero Dispersion Slope	: ≤ 0.092 ps/nm².km	Fibre Curl	: 4 meter radius of curvature

Cable Mechanical & Physical Characteristics

Cable Mechanical Characteristics		**Cable Physical Characteristics**	
Max. Tensile Strength	: 350 N (Short Term)	Cable Diameter	: 5.8mm / 6mm / 6.5mm / 7mm
Crush Resistance	: 1000 N/10 CM (Short Term)	Nominal Cable Weight	: Approx. 28 kg/km
Impact Resistance	: 10 N 0.5 m	Packing Length	: 1/2 km ± 5%
Min. Bending Radius (Long Term)	: 20D (D= Cable Diameter mm)	Packing Details	: Coil Form

All Tests Shall be Carried out as per IEC Standards. Change in attentuation Shall be ≤0.05 dB/km

Cable Printing Details: (Hot Foil White Emb.)	OFC wwY OPTRONIX PLATINUM xxF DC CFRP YW12 UT SM G652D Month YYYY Length Code Meter Marking Note: ww will be replaced with outer diameter of 58, 60, 65, 70. xx will be replaced with Fiber Count 02, 04, 06, 12.